

SONAR-X

Physics-Aware Explainable Marine Debris Intelligence System

SURVEY INTELLIGENCE REPORT

Survey Information

Survey ID	SX-SRV-20260925-004
Survey Name	SONAR Survey — Screenshot 2026 09 25 000056 — 25 Sep 2026
Survey Date	2026-09-25
Area	Not specified
Operator	Not specified
Data Source	SURVEY
Status	COMPLETED
Generated	2026-09-24 19:09 UTC
Software	SONAR-X 0.1.0-prototype

Sonar Metadata

Altitude above seabed	Not available
Range scale	Not available
Channel layout	BOTH
Already gain corrected	No
Already slant-range corrected	No

Detection Summary

Total detections	10
Marine Debris	1
Natural Formation	8
Sonar Artifact	0
Unknown Anomaly	1
Hazard — High Caution	1
Hazard — Caution	1

Detection List

Detection UID	Class	Conf.*	Evidence*	Risk	Hazard	Lat	Lon	Status
SX-DET-SX-SRV-20260925-004-010	UNKNOWN ANOMALY	53	53	MEDIUM	CAUTION	N/A	N/A	PENDING

Detection UID	Class	Conf.*	Evidence*	Risk	Hazard	Lat	Lon	Status
SX-DET-SX-SRV-20260925-004-009	MARINE DEBRIS	70	70	LOW	HIGH_CAUTION	N/A	N/A	PENDING
SX-DET-SX-SRV-20260925-004-008	NATURAL FORMATION	50	50	MEDIUM	NONE	N/A	N/A	PENDING
SX-DET-SX-SRV-20260925-004-007	NATURAL FORMATION	57	57	MEDIUM	NONE	N/A	N/A	PENDING
SX-DET-SX-SRV-20260925-004-006	NATURAL FORMATION	50	50	MEDIUM	NONE	N/A	N/A	PENDING
SX-DET-SX-SRV-20260925-004-005	NATURAL FORMATION	56	56	MEDIUM	NONE	N/A	N/A	PENDING
SX-DET-SX-SRV-20260925-004-004	NATURAL FORMATION	48	48	MEDIUM	NONE	N/A	N/A	PENDING
SX-DET-SX-SRV-20260925-004-003	NATURAL FORMATION	48	48	MEDIUM	NONE	N/A	N/A	PENDING
SX-DET-SX-SRV-20260925-004-002	NATURAL FORMATION	54	54	MEDIUM	NONE	N/A	N/A	PENDING
SX-DET-SX-SRV-20260925-004-001	NATURAL FORMATION	58	58	MEDIUM	NONE	N/A	N/A	PENDING

* Confidence and Evidence Score are prototype scores, NOT validated probabilities.

HAZARD ASSESSMENT

The following detections triggered precautionary hazard flags. This system does NOT identify specific hazardous objects. "Potential Hazard — Do Not Disturb" means sonar evidence warrants human review. Absence of a hazard flag does NOT mean an object is safe.

● SX-DET-SX-SRV-20260925-004-010 — CAUTION — UNKNOWN_ANOMALY

Indicators: Regular / structured geometry, Limited metadata / reduced signal quality

● SX-DET-SX-SRV-20260925-004-009 — HIGH_CAUTION — MARINE_DEBRIS

Indicators: Regular / structured geometry, Prominent acoustic shadow, Limited metadata / reduced signal quality

Limitations and Disclaimers

- This is a prototype system. All detections require human verification before any action.
- Confidence and evidence scores are prototype heuristic values, NOT validated probabilities.
- Side-scan sonar does not directly provide water temperature or depth.
- Conventional side-scan sonar is not a direct bathymetric sensor.
- Slant-range correction geometry is approximate where metadata was unavailable.
- Geolocation is approximate. GPS coordinates require field calibration.
- A missing detection does NOT prove debris was removed or moved.
- Absence of a hazard flag does NOT mean an object is safe.

AI-generated observations require verification by qualified human experts.